Here are the detailed, standalone notes from the video "p-hacking: What it is and how to avoid it!"

Video: p-hacking: What it is and how to avoid it!

Link: https://youtu.be/HDCOUXE3HMM

1. What is p-hacking?

P-hacking is the "misuse and abuse of analysis techniques" that results in being fooled by **false positives** [02:17].

It's a collection of bad scientific practices where a researcher, consciously or unconsciously, makes decisions *during* the analysis that are designed to produce a "statistically significant" p-value (typically $p < 0.05$). It's a way of "cheating" to find a positive result, even when one doesn't really exist.

2. The Core Problem: The 1-in-20 Rule (False Positives)

To understand p-hacking, you must first understand the p-value significance threshold (α).

- **The Threshold ($\alpha = 0.05$):** When we set our significance level to 0.05, we are accepting a 5% (or 1-in-20) chance of a **false positive** [06:28].
- **False Positive:** This is when you "reject the null hypothesis" (i.e., you conclude there *is* a difference) when in reality, the **null hypothesis was true** (there is no difference, and your result was a fluke) [06:18].
- **The Statistical Fact:** If you run 20 tests on things that have *no real effect* (like 20 different sugar pills), you should *expect* one of them to produce a p-value < 0.05 *purely by random chance* [06:47].

P-hacking involves any technique that exploits this 1-in-20 chance to make a random result look like a real discovery.

3. Type 1 of p-hacking: The Multiple Testing Problem ("Cherry-Picking")

This is the most common and obvious form of p-hacking.

- **The P-Hack:** A researcher wants to find a drug to cure a virus. They create 20 different candidate drugs (A, B, C... T) and test them all against a control group [00:36].
- **The "Discovery":** 19 of the drugs (A-D, F-T) show no effect (e.g., $p = 0.8, p = 0.3$, etc.). But one drug, **Drug E**, just by random chance, shows a p-value of 0.02 [01:43].
- **The "Sin":** The researcher **"cherry-picks"** this one result [08:43]. They throw away the 19 failed experiments and write a paper *only* about Drug E, claiming it's a success. They have presented a 1-in-20 random chance event as a real scientific finding. This is also known as the **Multiple Testing Problem** [07:15].

How to Avoid This:

- **The Solution:** You must **adjust for multiple tests**.
- **The Method:** The video names the **False Discovery Rate (FDR)** as a popular method [07:29].
- **How it Works:** You must input the p-values from *all* the tests you ran (all 20, not just the "good" one) [08:30]. The FDR algorithm then adjusts these p-values to account for the high probability of false positives.
- **The Result:** The "significant" $p = 0.02$ for Drug E will be adjusted to a much larger, non-significant value (e.g., $p = 0.4$), correctly revealing it as a false positive [08:04].

4. Type 2 of p-hacking: Optional Stopping ("Just Adding More Data")

This is a less obvious, but equally harmful, form of p-hacking.

- **The Scenario:** A researcher runs an experiment and gets a p-value that is "close" but not significant, for example, $p = 0.06$ [09:04].
- **The P-Hack:** Disappointed, the researcher thinks, "I'm so close! I'll just add one more measurement to each group" [09:49]. They re-calculate the p-value.
  - Maybe it's now $p = 0.07$. "Still not there... I'll add another."
  - They do this again, and now it's $p = 0.049$. "Hooray! It's significant! I'll stop collecting data and publish!" [09:59].
- **The "Sin":** This is called **optional stopping**. The statistical theory of p-values assumes you *only calculate one p-value* at the very end of an experiment [01:10:43]. By repeatedly checking the p-value after each new data point, you are simply giving random chance more and more opportunities to "walk" below the 0.05 line, creating a false positive [01:10:32].

How to Avoid This:

- **The Solution:** You must determine your **sample size *before*** you begin the experiment [01:11:31].
- **The Method:** This is done using a **Power Analysis** [01:11:41].
- **How it Works:** A power analysis tells you how many samples (e.g., people per group) you need to have a high probability of *correctly* rejecting the null hypothesis (i.e., finding a real effect, if one exists).
- **The Rule:** You must run your power analysis, determine your sample size (e.g., "I need 50 people per group"), and *then* collect all 100 data points. Only after all data is collected are you allowed to run your **one** statistical test [01:12:55]. If $p = 0.06$ at that point, that is your final answer.

Summary: How to Avoid p-hacking

1. **If you test many different things:** You *must* calculate a p-value for every single test, and then use a correction method like the **False Discovery Rate (FDR)** to adjust all your p-values [01:12:30].
2. **If you get a "close" p-value (like 0.06):** You **cannot** add more data and re-test. The experiment is over. The correct next step is to use your current data to run a **power analysis** to design a *new, larger, and properly-powered* experiment [01:12:55].
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